Problem 12 | Problems T5_MechanicalProperties.pdf #12 | True stress · true strain

Steel Cable: True Fracture Stress and True Strain from the Reduction in Area

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12. A steel cable has fracture strength of 300 MPa. Its ductility is 77% measured as a reduction in area. Calculate the real fracture strength. Considering that strain is uniform until fracture, what is the real strain?

Official Solution: $\varepsilon_{real} = 1.47$

HYPOTHESES & PARAMETERS

Hypotheses

1. **Fracture strength = engineering stress at fracture.** The "fracture strength of 300 MPa" is the load at fracture divided by the *initial* area, $\sigma_f = F_f / A_0$, as in the engineering curve of the tensile test [Session 11 Slide 9]. The true fracture stress is that load divided by the area at fracture, A_f [Session 11 Slide 37].
2. **Ductility as reduction in area.** $RA\,(\%) = \dfrac{A_0 - A_f}{A_0} \times 100$ is measured after fracture [Session 11 Slide 33], so $RA = 77\,\%$ means $A_f = 0.23\,A_0$.
3. **Uniform deformation.** The statement asks to take the strain as uniform until fracture, so the volume of the wire is constant, $l_0 A_0 = l_f A_f$ (the necking is not modelled) [Session 11 Slide 38].
4. **Real strain by integration.** $\varepsilon_r = \int_{l_0}^{l} dl/l = \ln(l/l_0)$ [Session 11 Slide 37].

Data

Quantity	Symbol	Value	In SI
Engineering fracture strength	σ_f	300 MPa	300×10^6 Pa
Reduction in area	RA	77 %	0.77 (dimensionless)

Unknowns: the true fracture stress $\sigma_{f,r}$ and the true strain at fracture ε_r.

MODEL & STRATEGY

One direction (the axis of the wire); no change of basis.

Strategy.

1. The reduction in area gives the ratio $A_f/A_0 = 1 - RA$.
2. The load at fracture is the same in both definitions of stress, $F_f = \sigma_f A_0 = \sigma_{f,r} A_f$, so $\sigma_{f,r} = \sigma_f\, A_0/A_f$ (the true stress is larger than the engineering one because the area has decreased) [Session 11 Slide 38].
3. With constant volume, the length ratio is the inverse of the area ratio, and the true strain follows from $\ln(l/l_0) = \ln(A_0/A_f)$.
4. The result is verified with the other route, $\sigma_r = \sigma(1+\varepsilon)$ and $\varepsilon_r = \ln(1+\varepsilon)$, through the engineering strain.

STEP-BY-STEP CALCULATION

1. Area ratio at fracture

$$\frac{A_f}{A_0} = 1 - \frac{RA}{100} = 1 - 0.77 = 0.23$$

2. True fracture stress

The load at fracture is $F_f = \sigma_f A_0$. Dividing by the area at fracture instead of the initial one:

$$\sigma_{f,r} = \frac{F_f}{A_f} = \frac{\sigma_f A_0}{A_f} = \frac{\sigma_f}{A_f/A_0} = \frac{300\ \text{MPa}}{0.23} = \mathbf{1304.3\ MPa}$$

($\sigma_{f,r}/\sigma_f = 4.348$: the true stress is more than four times the engineering one.)

3. True strain at fracture

With the volume constant, $l_0 A_0 = l_f A_f$, so

$$\frac{l_f}{l_0} = \frac{A_0}{A_f} = \frac{1}{0.23} = 4.348$$

and the integral of the definition of the real strain, with the substitution $u = l$, $du = dl$, and Barrow's rule between l_0 and l_f, is

$$\varepsilon_r = \int_{l_0}^{l_f} \frac{dl}{l} = \left[\ln l\right]_{l_0}^{l_f} = \ln l_f - \ln l_0 = \ln \frac{l_f}{l_0} = \ln \frac{A_0}{A_f} = \ln \frac{1}{0.23} = \mathbf{1.4697}$$

Equivalently, in terms of the diameters of the wire, $\varepsilon_r = 2\ln(D_0/D_f)$ [Session 11 Slide 38], with $D_f/D_0 = \sqrt{A_f/A_0} = \sqrt{0.23} = 0.4796$: $2\ln(1/0.4796) = 1.4697$.

4. Check through the engineering strain

The engineering strain at fracture is $\varepsilon = l_f/l_0 - 1 = 4.348 - 1 = 3.348$ (that is, 334.8 %). Then

$$\sigma_{f,r} = \sigma_f\,(1+\varepsilon) = 300 \times 4.348 = 1304.3\ \text{MPa} \qquad \varepsilon_r = \ln(1+\varepsilon) = \ln 4.348 = 1.4697$$

in agreement with the two results above.

RESULT & CHECKS

Checks

- **Units.** Stress in MPa; A_f/A_0 and the strains are dimensionless; the argument of the logarithm is a ratio of areas.
- **Limits.** If $RA \to 0$ (no necking) then $A_f \to A_0$, $\sigma_{f,r} \to \sigma_f$ and $\varepsilon_r \to 0$. As $RA \to 100\,\%$, both $\sigma_{f,r}$ and ε_r grow without bound.
- **Ordering.** $\sigma_{f,r} > \sigma_f$ and $\varepsilon_r < \varepsilon$ ($1.4697 < 3.348$), as stated in [Session 11 Slide 37].

Interpretation

The steel cable breaks when the material that is left carries 1304 MPa, not 300 MPa: the engineering value hides the reduction of the section. The true strain, 1.47, is the natural measure of such a large deformation, because the integral adds up the increments dl/l over the changing length.

LIMITATION OF THE UNIFORM-STRAIN HYPOTHESIS

A real cable necks before fracture, and most of the strain is concentrated in the neck. The mean strain along the whole length, obtained here, is therefore a lower bound of the local strain at the fracture section. The statement asks explicitly for the uniform strain, so that is the value reported.

Check against the official solution

Item	Official	Computed	Comment
true fracture stress	not given	1304.3 MPa	asked by the statement, absent from the key
true strain	1.47	1.4697	agrees

DISCREPANCY WITH THE OFFICIAL SOLUTION: THE KEY OMITS THE TRUE STRESS

The key only prints "$\varepsilon_{real} = 1.47$", although the statement asks for the real fracture strength first. The value computed here, $\sigma_{f,r} = 1304.3$ MPa, is not contradicted by the key: it is the first half of the answer that the key leaves out.